PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Vision Impairment and Cognitive Decline Among Older Adults: A Systematic Review
AUTHORS	Nagarajan, Niranjani; Assi, Lama; Varadaraj, V; Motaghi, Mina; Sun, Yi; Couser, Elizabeth; Ehrlich, Joshua; Whitson, Heather; Swenor, Bonnielin

VERSION 1 – REVIEW

REVIEWER	Akio Tada Hyogo Univ
REVIEW RETURNED	01-Feb-2021

GENERAL COMMENTS	Overview This study reviewed the association between vision impairment and cognitive decline among older adults literally. Abstract P3 line 19-21 “which may inform further research”: I cannot understand what does this sentence means. P3 line 29 “performed in Embase, ・・・” should be “ performed with Embase・・. Line 33-38 2 “The review included publications・・・” this sentence should be properly organized. It should address inclusion criteria. P4 line 8-10 “VI was associated with cognitive impairment.” The trend of association should be specified. Did those with VI have more cognitive decline? How about dementia? Methods P8 Data extraction must be done by more than 2 authors with approval. Results P8 line “Reasons for exclusion of studies were・・・・“ This part should be written in Methods section as exclusion criteria. Discussion In the discussion section, most of description are findings of studies reviewed and little discussion is found. Findings should be written in Results section. The authors should analyze trends for whether outcomes from different measurement could be differential from studies reviewed. How about the association by study design or measurement? The authors also should refer to bias of country where study was conducted. Conclusion P17 line 31-35 This sentence is not logical. Why “this overall

	agreement in the literature" suggests necessity of additional research? In the Conclusion section, what should be described in the Discussion section is found.

REVIEWER	Lei Zhang Wenzhou Medical University, Orthopaedics
REVIEW RETURNED	02-Feb-2021

GENERAL COMMENTS	This manuscript aimed to investigate the relationship between vision impairment and cognitive decline among older adults. The novelty is fair. First, a lot of jobs had been done to exclude more than 10 thousand literatures. We wonder how to ovid selection bias. Second, no RCTs are included and we wonder the level of evidence of the literatures. Finally, many patients with less than 60 years old were included. It cannot definitely imply the conclusion. Therefore, we don't suggest to publish the manuscript.

REVIEWER	Piers Dawes The University of Manchester
REVIEW RETURNED	11-Mar-2021

GENERAL COMMENTS	Thank you for the invitation to review this well-written and interesting systematic review. Minor suggestions: *this is mentioned in the discussion, but I suggest that the authors add to the intro (p6) the possibility that VI is a marker of risk for poor cognitive outcomes due to shared disease/aging processes. This would be important, because if true it means one could use VI as a marker of health/frailty risk that could signal a need for intervention. *p15. The most commonly used cognitive outcome measure was the MMSE. The MMSE and similar tests are dependent on good sensory function, so may be confounded with VI. MMSE may not be entirely valid and reliable (as asserted in the review) in the context of assessing people with VI. See Pye, A., Charalambous, A. P., Leroi, I., Thodi, C., & Dawes, P. (2017). Screening tools for the identification of dementia for adults with age-related acquired hearing or vision impairment: a scoping review. Int Psychogeriatr, 29(11), 1771-1784. Consider mentioning alternative cognitive assessments that do not rely on vision function (and perhaps stratifying the review results by VI-dependent/independent measures, if appropriate) *not sure if there would be scope to explore this in the analysis, but would be useful to note that implications of i) differences in pathology that underlie dementia subtypes and ii) differences in vision pathology - what are the implications for research linking any sort of VI with 'all cause' dementia? *some associations between certain sorts of VI and dementia may be due to shared pathology - see: Albers, M. W., Gilmore, G. C., Kaye, J., Murphy, C., Wingfield, A., Bennett, D. A., ... & Zhang, L. I. (2015). At the interface of sensory and motor dysfunctions and Alzheimer's disease. Alzheimer's & Dementia, 11(1), 70-98. *would it be useful to mention research using retinal biomarkers of

	dementia risk? *mention (in discussion) research that suggests a possible positive impact of vision interventions on cognitive outcomes, eg Maharani, A., Dawes, P., Nazroo, J., Tampubolon, G., Pendleton, N., & SENSE-Cog WP1 group. (2018). Cataract surgery and age-related cognitive decline: A 13-year follow-up of the English Longitudinal Study of Ageing. PLoS One, 13(10), e0204833.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Akio Tada, Hyogo Univ

Comments to the Author:
Overview
This study reviewed the association between vision impairment and cognitive decline among older adults literally.

Abstract
P3 line 19-21 “which may inform further research”: I cannot understand what does this sentence means.

Thank you for the comment. We have clarified this further as follows:

Page 2, line 6

“The objective of this systematic review is to synthesize the published literature on the association of VI with cognitive decline, cognitive impairment, or dementia, to aid the development of interventions, and guide public policies pertaining to the relationship between vision and cognition”

P3 line 29 “performed in Embase, ・・・” should be “ performed with Embase・・.

Made the change as per your suggestion,

Refer to Page 2, line 12:

“A literature search was performed with Embase, Medline, and Cochrane library databases from inception to March 2020, and included abstracts and articles published in peer-reviewed journals in English”

Line 33-38 2 “The review included publications・・・” this sentence should be properly organized. It should address inclusion criteria.

Thank you for the comment. Made the change as per your suggestion

Refer to Page 2, line 14:

“Our inclusion criteria included publications that contained subjective and/or objective measures of vision and cognition, or a diagnosis of VI, cognitive impairment or dementia”

P4 line 8-10 “VI was associated with cognitive impairment.” The trend of association should be specified. Did those with VI have more cognitive decline? How about dementia?

To clarify this further, we have made the following change:

Page 3, Line 8

“Our systematic review indicates that a majority of studies examining the vision-cognition relationship report that VI is associated with more cognitive decline, cognitive impairment, or dementia among older adults”

Methods
P8 Data extraction must be done by more than 2 authors with approval.
To address the risk of bias, a 10% sample of data extraction was done by the second reviewer (LA). Any discrepancies were addressed and adjudicated by the third reviewer (VV). We have clarified this further as follows:

Page 7, Line 6

“Data were extracted from the included publications by one author (NN), and another (LA) extracted data from a random sample of 10% to compare results. Any discrepancies were adjudicated by a third author (VV)”

Results
P8 line “Reasons for exclusion of studies were・・・・“ This part should be written in Methods section as exclusion criteria.

Made the change as per your suggestion. Please see page 6, line 11 in methods:

“Reasons for exclusion of studies were: 1. Outcome measure was not vision or cognition, 2. Association between vision and cognition was not explored, 3. Sample size <100, 4. Publication not in English, 5. Mean age <50 years, 6. No cognitive measure, 7. No vision measure, 8. Outcome was not part of inclusion criteria”

Discussion
In the discussion section, most of description are findings of studies reviewed and little discussion is found. Findings should be written in Results section.
The authors should analyze trends for whether outcomes from different measurement could be differential from studies reviewed.
How about the association by study design or measurement?

The authors also should refer to bias of country where study was conducted.

Page 13 line 19 mentions about the different conclusions seen from two different studies using the same dataset and attributes the findings to different outcomes measured (dementia vs cognitive impairment):

"Brenowitz et al [95] in their longitudinal study using the Health ABC data concluded that VA and CS independently were not significantly associated with incident dementia (VA: HR= 1.26, 95% CI 0.90-1.77, CS: HR=1.11 95% CI 0.88-1.38). However, Swenor et al [123] used data from the same study and found that impaired visual acuity, contrast sensitivity, and stereo acuity had a greater risk of incident cognitive impairment (VA impairment HR=1.55, 95% CI 1.12, 2.14 vs No VA impairment). This could be due to the different outcome measures assessed, i.e. dementia [95] vs cognitive impairment[123]"

To address the comment, we have looked at the VI-cognition association by study design.

Refer to Page 13, Line 7:

"Of the 91 studies that found an association between VI and cognitive function, 35 were longitudinal, 46 were cross-sectional and 10 were case-control studies"

In terms of measurement type, since there was great heterogeneity, it's not feasible to look at association by measurement type and we have addressed this in the discussion section:

Page 14, Line 8

"There was considerable heterogeneity in the measurement and reporting of cognitive function. Studies measured cognitive function using a variety of instruments with the most common being MMSE"

We have made a note regarding bias associated with studies from different countries under limitations section:

Refer to Page 16, Line 5

"There is also potential bias associated with studies that used different protocols for cognitive and sensory measurements"

Conclusion
P17 line 31-35 This sentence is not logical. Why "this overall agreement in the literature" suggests necessity of additional research?

In the Conclusion section, what should be described in the Discussion section is found.

Our review of the literature found strong consensus that vision impairment was associated with cognitive decline, impairment or dementia. We intend for this review to pave the way for additional

research in this space. Research should move away from documenting this relationship and instead understanding the mechanisms underlying this relationship which remains unclear

We have clarified this further:

Page 17, line 2

“This overall agreement in the literature suggests that poor visual and cognitive function are associated, and that additional research is now needed to move beyond documenting these associations”

Reviewer: 2
Dr. Lei Zhang, Wenzhou Medical University

Comments to the Author:
This manuscript aimed to investigate the relationship between vision impairment and cognitive decline among older adults. The novelty is fair.

First, a lot of jobs had been done to exclude more than 10 thousand literatures. We wonder how to avoid selection bias.

Second, no RCTs are included and we wonder the level of evidence of the literatures.

Finally, many patients with less than 60 years old were included. It cannot definitely imply the conclusion.

Therefore, we don't suggest to publish the manuscript.

Thank you for the comments. Our initial abstract screening process was performed by 4 different reviewers, where 2 reviewers each screened an abstract. We used Covidence which is an online software that is designed for and recommended to perform systematic review to help minimize issues of selection bias. Any discrepancies during abstract screening was adjudicated by a member of the other team. We followed Cochrane guidelines for performing systematic review during each step to minimize errors and bias.

There are no RCTs included because no RCTs exist. Our review was focused on association between vision impairment and cognitive function and did not test for any interventions.

The studies we included in our review consisted of longitudinal studies, which had patients aging into the cohort. Since our initial inclusion criteria was to include studies with participants aged 50 and above, we believe we have captured enough studies that have older adults in our population sample. The mean age of our population as noted in Table 1 is 73.0 years.

Reviewer: 3
Dr. Piers Dawes, The University of Manchester

Comments to the Author:
Thank you for the invitation to review this well-written and interesting systematic review.

Minor suggestions:
*this is mentioned in the discussion, but I suggest that the authors add to the intro (p6) the possibility that VI is a marker of risk for poor cognitive outcomes due to shared disease/aging processes. This would be important, because if true it means one could use VI as a marker of health/frailty risk that could signal a need for intervention.

Thank you for the comment. We have clarified this as follows:

Page 5, Line 8

"Prior work has suggested that cognition and vision are associated,[6, 7] and while there are shared risk factors (neuropathological/vascular),[8] there is also longitudinal evidence that VI is associated with cognitive changes"

See reference number:8

*p15. The most commonly used cognitive outcome measure was the MMSE. The MMSE and similar tests are dependent on good sensory function, so may be confounded with VI. MMSE may not be entirely valid and reliable (as asserted in the review) in the context of assessing people with VI. See Pye, A., Charalambous, A. P., Leroi, I., Thodi, C., & Dawes, P. (2017). Screening tools for the identification of dementia for adults with age-related acquired hearing or vision impairment: a scoping review. Int Psychogeriatr, 29(11), 1771-1784.
Consider mentioning alternative cognitive assessments that do not rely on vision function (and perhaps stratifying the review results by VI-dependent/independent measures, if appropriate)

Thank you for this valuable comment. We are cognizant of the fact that cognitive assessment relies heavily on sensory function and our team is currently working on a paper that addresses this issue. We surveyed cohort studies collecting cognitive data among older adults and observed significant variation in exclusion criteria related to hearing or vision impairment, methods of assessment, as well as accommodation practices.

We agree with the reviewer that there is value in examining the association between vision and cognition based on cognitive tests that do and do not rely on vision. However, it is beyond the scope of this review. We have added this to our limitation and cited the above papers:

Page 16, Line 5

"There is also potential bias associated with studies that used different protocols for cognitive and sensory measurements. The MMSE, which was the most commonly used assessment method for testing cognition is sensory dependent and therefore one can argue that the results may be confounded with vision impairment.[128] Further studies should examine the impact of using vision independent cognitive tests on the vision-cognition relationship."

See reference 128

*not sure if there would be scope to explore this in the analysis, but would be useful to note that implications of i) differences in pathology that underlie dementia subtypes and ii) differences in vision pathology - what are the implications for research linking any sort of VI with 'all cause' dementia?

Thank you for this valuable comment. Unfortunately, this analysis is beyond the scope of our systematic review and thus have added this to our limitation section as follows:

Page 16, line 10

“Our review examined all cause vision impairment and dementia, and further study is needed to examine the vision-cognition relationship by dementia subtype and by different vision pathology”

Our team is currently working on examining the relationship between VI and dementia subtype using Baltimore Longitudinal Study of Aging (BLSA) data.

*some associations between certain sorts of VI and dementia may be due to shared pathology - see: Albers, M. W., Gilmore, G. C., Kaye, J., Murphy, C., Wingfield, A., Bennett, D. A., ... & Zhang, L. I. (2015). At the interface of sensory and motor dysfunctions and Alzheimer's disease. Alzheimer's & Dementia, 11(1), 70-98.

Thank you for the comment and citation. We have mentioned this in the introduction section, and I have added the citation to Page 5, line 9. See reference number: 8

*would it be useful to mention research using retinal biomarkers of dementia risk?

Anatomical retinal biomarkers are different from functional vision impairment as assessed in this paper. We have collected information on studies that have used Optical Coherence Tomography (OCT) and plan to submit a separate manuscript with our findings as a follow-up to this paper.

*mention (in discussion) research that suggests a possible positive impact of vision interventions on cognitive outcomes, eg
Maharani, A., Dawes, P., Nazroo, J., Tampubolon, G., Pendleton, N., & SENSE-Cog WP1 group. (2018). Cataract surgery and age-related cognitive decline: A 13-year follow-up of the English Longitudinal Study of Ageing. PLoS One, 13(10), e0204833.

Thank you for the comment. I have made a note as follows:

Page 15, Line 1 and cited the above paper.

“Literature also suggests that vision rehabilitation in the form of cataract surgery slows the rate of cognitive decline, and therefore, early vision interventions could potentially reduce risk of dementia” see reference 127

VERSION 2 – REVIEW

REVIEWER	Akio Tada Hyogo Univ
REVIEW RETURNED	02-Jun-2021

GENERAL COMMENTS	Overview Improvements are seen in the present manuscript. However, there are some problems to be published. Specific points Although I mentioned that description of data should not be in the Discussion section in the previous review, it is not still improved. For example, p92 line 22-p94 line 10. These are just detailed descriptions of “the finding of the study”. We can see the figures concerning HR or OR, 95%CI and findings of each study in Table 3 and 4 unless they are written in the Discussion. Authors need to address points such as the followed: 1) What do the results mean for the problem and hypothesis of the study? 2) What are the implications for the target population? 3) How do the findings add to the current literature (which you would have included in the Background section when writing about the literature review)? 4) What are the gaps, if any, you found in your study after evaluating the results? 5) Are there any suggestions for future research? Authors need to give a short summary of findings and discuss about what is written in p92-p94 in terms of these points of view. Conclusion Page 98 line 6 “largely consistent”: Use of expression that means positive association is recommended. “is largely consistent” also includes consistently negative.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Dr. Akio Tada, Hyogo Univ

Comments to the Author:
Overview
Improvements are seen in the present manuscript. However, there are some problems to be published.

Specific points
Although I mentioned that description of data should not be in the Discussion section in the previous review, it is not still improved. For example, p92 line 22-p94 line 10. These are just detailed descriptions of “the finding of the study”. We can see the figures concerning HR or OR, 95%CI and findings of each study in Table 3 and 4 unless they are written in the Discussion.

As per your recommendation, we have removed the HR, OR and 95% CI from the discussion section. However, our reason to retain some of the text describing the studies is to highlight specific findings from studies that obtained different ratings from our quality assessment, as well as the different

measures of cognition and vision, that will help the reader interpret the results appropriately. Since the tables have a lot of information, we believe there is value in highlighting these specific findings that we would like to draw attention to.

Authors need to address points such as the followed: 1) What do the results mean for the problem and hypothesis of the study?

Thank you for this comment.

Refer to 11, Line 15

“Results from the longitudinal studies that found a positive association between vision and cognition supports our hypothesis that VI may be a risk factor cognitive impairment, cognitive decline, or dementia”

2) What are the implications for the target population?

To clarify this further, please refer to the implications section under discussion, Page 15 Line 10

“Third, as the longitudinal results support the sensory consequence theory, and suggest that VI may be a risk factor for cognitive decline, this points to a need for formulating preventive measures and vision rehabilitation models, such as prescription glasses, cataract surgery, low vision rehabilitation, etc., that could have the potential to improve overall health and well-being of older adults.”

3) How do the findings add to the current literature (which you would have included in the Background section when writing about the literature review)?

I have made the following changes to clarify this further.

Please refer to Page 11, Line 13

“In this systematic review, we evaluated and synthesized the literature examining the association between VI and cognitive function among older adults, and found strong agreement that VI is associated with cognitive impairment, cognitive decline, or dementia”

4) What are the gaps, if any, you found in your study after evaluating the results?

I have added the following in the limitations section to expand on this point further.

Please refer to Page 15, Line 20

“Although 35 longitudinal studies found a positive association between VI and cognitive decline, we cannot establish temporality between this relationship due to the heterogenous nature of the studies.”

5) Are there any suggestions for future research?

Please refer to Page 16, Line 19

“The focus of this area of research should now turn to identifying the factors and strategies that mediate the vision-cognition relationship and identifying potential interventions, such as vision rehabilitation models and strategies tailored to people with VI, that may mitigate the cognitive implications of VI”

Authors need to give a short summary of findings and discuss about what is written in p92-p94 in terms of these points of view.

Conclusion
Page 98 line 6 “largely consistent”: Use of expression that means positive association is recommended. “is largely consistent” also includes consistently negative.

To clarify this further, please refer to Page 16, Line 15

“This systematic review found that the positive association of VI with cognitive decline, cognitive impairment, or dementia is largely consistent across studies using different measures of vision and cognition, as well as between countries and cohorts”